Introductory Econometrics with Recitation: TA Session Week 14

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Today's agenda

1. Instrumental Variable (IV) Regression
   - Story and Model Setup
   - OLS and IV Model
2. IV Validity
   - Relevance
   - Exogeneity
3. Concluding Remark
   - Acknowledgement

Today's Dataset

- `smoke_ban.csv`: a cross-sectional dataset on smoking behavior and income from the `wooldridge` package.
- Import the data:

```
smoke <- wooldridge::smoke
```

- The main question is whether cigarette smoking affects income.

Variable	Description
`lincome`	Log of annual income
`cigs`	Number of cigarettes smoked per day
`cigpric`	State cigarette price, measured in cents per pack.
`restaurn`	Restaurant smoking ban; $1 =$ Ban, $0 =$ No ban
`educ`	Years of education
`age`	Age in years
`agesq`	Age squared

Story: Costly Cigarettes

- Smoking cigarettes is not only addictive, but also costly.
- We want to study the effect of smoking on income.
- Consider the following model:

$$lincome_i = \beta_0 + \beta_1 cigs_i + \beta_2 educ_i + \beta_3 age_i + \beta_4 agesq_i + u_i.$$

- The variable `cigs` may be endogenous, so β_1 may not have a causal interpretation.

Story: Costly Cigarettes

- Some states in the U.S. have implemented smoking bans in restaurants.
- This provides a possible instrumental variable: `restaurn`.
- The IV idea requires two conditions:
  - **Relevance:** restaurant bans affect smoking behavior.
  - **Exogeneity:** restaurant bans affect income only through smoking behavior.
- We use two-stage least squares (2SLS) to estimate the IV model.

Two-Stage Least Squares

- **First stage:**

$$cigs_i = \pi_0 + \pi_1 restaurn_i + \pi_2 educ_i + \pi_3 age_i + \pi_4 agesq_i + v_i.$$

- **Second stage:**

$$lincome_i = \beta_0 + \beta_1 \widehat{cigs}_i + \beta_2 educ_i + \beta_3 age_i + \beta_4 agesq_i + u_i.$$

- Here, $\widehat{cigs}_i$ is the predicted value from the first-stage regression.

OLS and IV Model

- **OLS model**

```
ols <- lm_robust(
  lincome ~ cigs + educ + age + agesq,
  data = smoke
)
```

- **IV model**

```
library(ivreg)

iv <- ivreg(
  lincome ~ cigs + educ + age + agesq |
    restaurn + educ + age + agesq,
  data = smoke
)
```

Model Comparison

- **Output:**

```
========================================
                 OLS           IV
----------------------------------------
(Intercept)     7.7954 ***    7.7811 ***
               (0.2085)      (0.2281)
cigs            0.0017       -0.0414
               (0.0014)      (0.0260)
educ            0.0604 ***    0.0400 *
               (0.0075)      (0.0162)
age             0.0577 ***    0.0932 ***
               (0.0092)      (0.0237)
agesq          -0.0006 ***   -0.0010 ***
               (0.0001)      (0.0003)
----------------------------------------
R^2             0.1650       -0.4941
Adj. R^2        0.1608       -0.5015
Num. obs.          807           807
RMSE            0.6529
========================================
*** p < 0.001; ** p < 0.01; * p < 0.05
```

IV Estimation with Robust SE

- Use coeftest() with a robust variance-covariance estimator.

```
library(lmtest)   # for coeftest()
library(sandwich) # for robust SE

coeftest(iv, vcov = sandwich)
```

- **Output:**

```
t test of coefficients:

               Estimate  Std. Error t value  Pr(>|t|)
(Intercept)  7.78113928  0.25650420 30.3353 < 2.2e-16 ***
cigs        -0.04138462  0.02403335 -1.7220  0.085461 .
educ         0.04002417  0.01536913  2.6042  0.009379 **
age          0.09320769  0.02230429  4.1789 3.251e-05 ***
agesq       -0.00104373  0.00025489 -4.0948 4.655e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Relevance: First-Stage F Test

- The instrument should be correlated with the endogenous variable.
- We check this using the first-stage regression.

```
first <- lm_robust(
  cigs ~ restaurn + educ + age + agesq,
  data = smoke
)

summary(first)
```

- A common rule of thumb is that the first-stage F-statistic should be larger than 10.

Relevance: First-Stage F Test

- **Output:**

```
... (omitted)

Multiple R-squared: 0.0418
Adjusted R-squared: 0.03822
F-statistic: 16.46 on 3 and 803 DF
p-value: 2.176e-10
```

- Since the F-statistic is larger than 10, the instrument is not weak by this rule of thumb.

Exogeneity: Overidentification Test

- If we have more instruments than endogenous variables, we can use the J test to test instrument exogeneity.
- After estimating the IV model, regress the 2SLS residual on the instruments:

$$\widehat{u}_i^{2SLS} = \gamma_0 + \gamma_1 Z_{1i} + \gamma_2 Z_{2i} + \cdots + \gamma_m Z_{mi} + \epsilon_i.$$

- The null hypothesis is that all instruments are exogenous:

$$H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_m = 0.$$

Exogeneity: Overidentification Test

- The test statistic is:
$$J = nR^2 \sim \chi^2_{m-k},$$
where m is the number of instruments and k is the number of endogenous variables.
- If we reject the null, then at least one instrument may be invalid.
- If we fail to reject the null, then the test does not find evidence against instrument exogeneity.
- This test requires overidentification, so it cannot be used when there is only one instrument for one endogenous variable.

J-Test in R

- To run a J-test, suppose we use two instruments: `restaurn` and `cigpric`:

```
iv_overid <- ivreg(
  lincome ~ cigs + educ + age + agesq |
    restaurn + cigpric + educ + age + agesq,
  data = smoke
)
summary(iv_overid, diagnostics = TRUE)
```

J-Test in R

- **Output:**

```
... (omitted)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.7812628  0.2272624  34.239  < 2e-16 ***
cigs        -0.0410122  0.0259272  -1.582 0.114085
educ         0.0401998  0.0160987   2.497 0.012721 *
age          0.0929009  0.0235875   3.939 8.91e-05 ***
agesq       -0.0010402  0.0002713  -3.834 0.000136 ***

Diagnostic tests:
                  df1 df2 statistic p-value
Weak instruments    2 801     3.131  0.0442 *
Wu-Hausman          1 801     4.887  0.0273 *
Sargan              1  NA     1.519  0.2177
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.87 on 802 degrees of freedom
Multiple R-Squared: -0.4827,^^IAdjusted R-squared: -0.4901
Wald test: 22.79 on 4 and 802 DF,  p-value: < 2.2e-16
```

Acknowledgement

- This is the last TA session, and next week will be office hour.
  - Thank you for joining the TA sessions this year!
  - I hope the TA sessions were helpful.
- If you have questions or just want to chat, my door is always open.
- You can reach me by email or in person.
  - Course suggestions
  - Econometrics and empirical research
  - Graduate school preparation and applications

Suggested Courses for Future Learning

- **Econometrics I and II**
  - Revisit concepts from this semester with vector and matrix notation.
  - Introduce more empirical methods and models.
  - Usually require an empirical research project.
- **Econometrics Theory I and II**
  - Extend econometric models using linear algebra.
  - Focus on both theory and applications.
  - Cover asymptotic theory, panel data, and potential outcomes.